CLASS XII PHYSICS — REVISION NOTES

Newtonian Mechanics: Oscillations & Simple Harmonic Motion (SHM)

BHSEC TRIAL EXAM SPECIAL

Teacher's Overview & Foundations: Newtonian mechanics operates in inertial frames assuming absolute distance, time, and mass. Oscillations surround us—from simple pendulums, piano strings, and heartbeats to acoustic sound waves, thermal atomic vibrations, radio transmissions, and earthquake seismic activity. Understanding SHM is essential across mechanical, electrical, and atomic physics.

CURRICULUM FRAMEWORK & OBJECTIVE MAPPING

1.1.1 & 1.1.2 Scope & Objectives:

- Differentiate Periodic & Oscillatory Motions.
- Derive SHM from Uniform Circular Motion (UCM).
- Determine equations for displacement (x), velocity (v), and acceleration (a).
- Graphical analysis using interactive simulations.

Competencies & Assessment Focus:

- Apply SHM in amusement parks, open-space gyms, and seismometers.
- Conduct CER (Claim-Evidence-Reasoning) for energy conservation in SHM.
- Investigate resonance via tuning forks and design earthquake-resistant models.

1. ESSENTIAL CONCEPTS & FUNDAMENTAL DEFINITIONS

- **Periodic Motion:** Motion repeating identically at equal intervals of time T (e.g., planetary orbits, hands of a clock).
- **Oscillatory (Vibrational) Motion:** To-and-fro periodic motion about a fixed mean position (e.g., mass-spring system, pendulum).

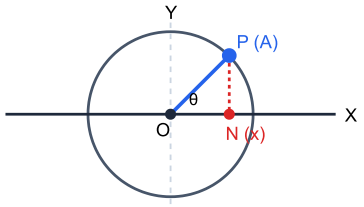
Exam Golden Rule: *Every simple harmonic motion is periodic, but every periodic motion need NOT be simple harmonic.* (Example: Uniform circular motion is periodic, but not SHM).

Simple Harmonic Motion (SHM): Oscillatory motion where restoring force F (and acceleration a) is directly proportional to displacement x from the mean position and acts in the opposite direction:

$$F = -k\,x \quad \text{or} \quad a = -\omega^2\,x$$

2. DERIVATION: SHM AS PROJECTION OF UNIFORM CIRCULAR MOTION

Board Exam Mandate: Show that the projection of uniform circular motion along a diameter executes SHM (using horizontal displacement x).



Step 1: Displacement Equation

Let a reference particle P move in a circle of radius A with constant angular velocity ω.
At $t = 0$, initial phase = ϕ. At time t, angle rotated $\theta = \omega t$.
Total angle with X-axis = $(\omega t + \phi)$.
Projecting P onto horizontal diameter (X-axis):

$$x(t) = A\cos(\omega t + \phi)$$

(Note: Projection on Y-axis gives $y(t) = A\sin(\omega t + \phi)$).

Step 2: Velocity in SHM

Differentiating displacement $x(t)$ with respect to time t:

$$v = dx/dt = d[A\cos(\omega t + \phi)]/dt = -A\,\omega\sin(\omega t + \phi)$$

Using identity $\sin\theta = \sqrt{(1 - \cos^2\theta)}$ and $\cos(\omega t + \phi) = x/A$:

$$v = -A\,\omega\,\sqrt{(1 - x^2/A^2)} = \pm\,\omega\,\sqrt{(A^2 - x^2)}$$

- **At Mean Position ($x = 0$):** Velocity is maximum: $v_{max} = \pm\,A\,\omega$
- **At Extreme Positions ($x = \pm A$):** Velocity is zero: $v = 0$

Step 3: Acceleration in SHM
Differentiating velocity $v(t)$ with respect to time t:

$$a = dv / dt = d[-A\,\omega\,sin(\omega t + \phi)] / dt = -A\,\omega^2\,cos(\omega t + \phi)$$

Since $x = A\,cos(\omega t + \phi)$:

$$a = -\omega^2\,x$$

Conclusion: Since $a \propto -x$ (acceleration is directly proportional to displacement and opposite in direction), the projection of uniform circular motion on a diameter executed as **Simple Harmonic Motion**.

- **At Mean Position ($x = 0$):** Acceleration is zero ($a = 0$).
- **At Extreme Positions ($x = \pm A$):** Acceleration is maximum: $a_{max} = \omega^2\,A$.

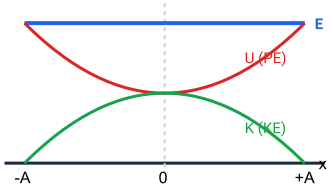
3. DERIVATION: ENERGY CONSERVATION IN SHM

BHSEC Focus Question: Derive expressions for Potential, Kinetic, and Total Mechanical Energy, and prove conservation.

1. **Potential Energy (U):** Work done against restoring force $F = -kx$ for displacement x:
   $U = \int_0^x k\,x\,dx = ½\,k\,x^2 = ½\,m\,\omega^2\,x^2$
2. **Kinetic Energy (K):** Energy due to motion:
   $K = ½\,m\,v^2 = ½\,m\,\omega^2\,(A^2 - x^2) = ½\,k\,(A^2 - x^2)$
3. **Total Mechanical Energy (E):**
   $E = U + K = ½\,k\,x^2 + ½\,k\,(A^2 - x^2)$

$$E = ½\,k\,A^2 = ½\,m\,\omega^2\,A^2 = Constant$$

CER Conclusion: Total energy depends only on mass m, frequency ω, and amplitude A. It is completely independent of displacement x and time t, proving total energy is conserved.



Energy vs Displacement (x)

4. KEY FORMULA SUMMARY FOR QUICK REVISION

Physical Quantity	Formula (in terms of x)	Key Exam Notes / Extreme Values
Time Period & Frequency	$T = 2\pi / \omega = 1 / f$	$f = (1 / 2\pi)\,\sqrt{(k / m)}$
Displacement	$x(t) = A\,cos(\omega t + \phi)$	Max at extreme ($x = \pm A$), Zero at mean ($x = 0$)
Velocity	$v = \pm\,\omega\,\sqrt{(A^2 - x^2)}$	$v_{max} = A\omega$ (at $x = 0$); $v = 0$ (at $x = \pm A$)
Acceleration	$a = -\omega^2\,x$	$a_{max} = \omega^2 A$ (at $x = \pm A$); $a = 0$ (at $x = 0$)
Simple Pendulum Period	$T = 2\pi\,\sqrt{(l / g)}$	Independent of mass of bob; slope of T^2 vs l gives g
Total Energy	$E = ½\,k\,A^2 = ½\,m\,\omega^2\,A^2$	Constant at all positions (conserved)

5. CORE SUB-TOPICS QUICK REVIEW (SECTIONS 1.1.3 – 1.1.5)

1.1.4 Simple Pendulum & Gravity (g):

- Restoring Torque / Force: $F = -mg\,sin\,\theta \approx -mg(x / l)$.
- Valid for small angle approximation ($sin\,\theta \approx \theta$).
- Time period $T = 2\pi\,\sqrt{(l / g)}$ is independent of bob mass or material.

1.1.5 Forced Oscillations & Resonance:

- **Free:** Oscillates at natural frequency f_0.
- **Forced:** Driven by external periodic force at frequency f.
- **Resonance:** When $f = f_0$, energy transfer is maximum, producing maximum amplitude surge.

6. SOLVED PAST BHSEC BOARD & TRIAL EXAM QUESTIONS

Question 1 (BHSEC Trial Exam — 4 Marks)

(a) Define simple harmonic motion (SHM).
(b) A body executes SHM with an amplitude of 0.05 m and a frequency of 10 Hz. Calculate its maximum velocity and maximum acceleration.

Solution:
(a) **Definition:** SHM is periodic to-and-fro motion about a fixed mean position where acceleration is directly proportional to displacement and acts in the direction opposite to displacement ($a \propto -x$).
(b) **Given:** $A = 0.05\ m, f = 10\ Hz$.
1. Angular frequency: $\omega = 2\pi f = 2 \times \pi \times 10 = 20\pi \approx 62.83\ rad/s$
2. Max Velocity: $v_{max} = A\,\omega = 0.05 \times 62.83 = \mathbf{3.14\ m/s}$
3. Max Acceleration: $a_{max} = A\,\omega^2 = 0.05 \times (62.83)^2 = \mathbf{197.39\ m/s^2}$

Question 2 (BHSEC Conceptual Question — 2 Marks)

Explain why a simple pendulum clock loses time when transported to the summit of a high mountain.

Solution:
Time period of a simple pendulum is $T = 2\pi\sqrt{(l/g)}$. At the summit of a mountain, acceleration due to gravity g decreases ($g \propto 1/R^2$). Since $T \propto 1/\sqrt{g}$, a smaller g increases time period T. Each swing takes longer, causing the clock to run slow and **lose time**.

Question 3 (BHSEC MCQ — 1 Mark)

A particle of mass m executes SHM with angular frequency ω. If its displacement from equilibrium is halved, the restoring force becomes:
A. Doubled B. Halved C. One-fourth D. Unchanged

Answer: B. Halved.
Reason: Restoring force $F = -m\,\omega^2\,x$. Since $F \propto x$, halving displacement x directly halves restoring force F.

Question 4 (Analytical Question — 3 Marks)

Find displacement x from mean position where Kinetic Energy equals Potential Energy.

Solution:
Set $K = U \Rightarrow \frac{1}{2}\,k\,(A^2 - x^2) = \frac{1}{2}\,k\,x^2$
$\Rightarrow A^2 - x^2 = x^2 \Rightarrow 2\,x^2 = A^2$
$\Rightarrow x = \pm A/\sqrt{2} \approx \mathbf{\pm\,0.707\,A}$

7. STUDENT TRIAL EXAM PRACTICE QUESTIONS (TRY THESE!)

1. **[1 Mark]** What is the time period of the oscillatory motion given by $f(t) = \cos(2\omega t + \phi)$?
2. **[2 Marks]** A girl is swinging on a swing in a sitting position. How will the time period change if she stands up? Explain using $T = 2\pi\sqrt{(l/g)}$.
3. **[2 Marks]** What effect would it have on the time period of a simple pendulum if its wooden bob were replaced with an identical iron bob of greater mass?
4. **[3 Marks]** A sewing machine needle moves in a path of length 4 cm and its frequency of oscillation is 10 Hz. Find its displacement and acceleration $t = 1/120\ s$ after crossing the center of its path.
5. **[4 Marks]** Explain the phenomenon of **resonance** with two real-life applications. How is resonance mitigated when designing earthquake-resistant infrastructures?